FYUGP B.Sc. MATHEMATICS HONOURS

ABSTRACT ALGEBRA I (MAT5CJ302)

Semester V · Core in Major

MODULE 4: HOMOMORPHISMS, RINGS, FIELDS AND INTEGRAL DOMAINS

Past questions

Section A: 1 to 3 marks

Homomorphisms and kernels

1 What is the necessary condition for a homomorphism ϕ from a group G to G' to be injective. **(1 mark)** *November 2018*

2 Let $\Phi : G_1 \to G_2$ be a group homomorphism. Show that $\Phi(e) = e'$ where e and e' are identity elements of G_1 and G_2 respectively. **(3 marks)** *November 2021*

Rings, subrings and units

3 What is the characteristic of the ring $\langle \mathbb{Z}_9, +_9, \times_9 \rangle$. **(1 mark)** *November 2018*

4 Give an example for an integral domain which is not a field. **(1 mark)** *November 2018*

5 State True or False. $\mathbb{Z}$ is a sub field of $\mathbb{Q}$. **(1 mark)** *November 2021*

6 Define a Ring. **(3 marks)** *November 2021*

7 Define commutative ring. Give an example. **(2 marks)** *November 2022*

8 Check whether the set $\{m + n\sqrt{2} \mid m, n \in \mathbb{Z} \text{ and } n \text{ is even}\}$ is a subring of the field of real numbers. **(2 marks)** *November 2023*

9 Define units in a ring. What are the units in $\mathbb{Z}$? **(2 marks)** *November 2025*

Zero divisors, integral domains and fields

10 How many zero divisors are there for the field $\mathbb{Z}_7$? **(1 mark)** *November 2021*

11 How many units are there for the field $\mathbb{Z}_7$? **(1 mark)** *November 2021*

12 Give an example of integral domain which not a field. **(1 mark)** *November 2021*

13 Give example of an integral domain. **(3 marks)** *November 2021*

14 Define Integral Domain. Give an example. **(2 marks)** *November 2022*

Section B: 4 to 7 marks

Homomorphisms and kernels

15 Define group homomorphism and state fundamental theorem of homomorphism. **(4 marks)** *November 2018*

16 Consider the rings $\langle \mathbb{Z}, +, \cdot \rangle$ and $\langle 2\mathbb{Z}, +, \cdot \rangle$. Verify whether the map $\phi : \mathbb{Z} \to 2\mathbb{Z}$ defined by $\phi(x) = 2x \ \forall x \in \mathbb{Z}$ is a ring homomorphism or not. **(4 marks)** *November 2021*

17 Let G_1 and G_2 be groups. Let $\phi : G_1 \to G_2$ be a group homomorphism with $\operatorname{Ker} \phi = \{x \in G_1 : \phi(x) = e\}$. Show that ϕ is one to one if and only if $\operatorname{Ker} \phi = \{e\}$. **(5 marks)** *November 2022*

18 Let G be an abelian group, and let n be any positive integer. Show that the function $\phi(x) = x^n$ is a homomorphism. **(5 marks)** *November 2022*

19 Let $\phi : G_1 \to G_2$ be a group homomorphism, with $K = \ker \phi$. Prove that K is a subgroup of G_1. **(5 marks)** *November 2023*

20 Let $\phi : G_1 \to G_2$ be a group homomorphism. If H_1 is a sub-group of G_1, then show that $\phi(H_1)$ is a sub-group of G_2. **(5 marks)** *November 2025*

The fundamental homomorphism theorem

21 State and prove fundamental homomorphism theorem. **(5 marks)** *November 2023*

22 State and prove First Isomorphism Theorem. **(5 marks)** *November 2025*

Rings, subrings and units

23 Give an example of non-commutative finite ring. Establish that it is so. **(4 marks)** *November 2018*

24 Find a formula for identifying units in the ring of guassian integers $\{a + ib : a, b \in \mathbb{Z}\}$. **(4 marks)** *November 2018*

25 Show that any subring of a field is an integral domain. **(5 marks)** *November 2022*

Zero divisors, integral domains and fields

26 Define a field. Show that $(a + b)^2 = a^2 + 2ab + b^2$ in any field using the axioms of the field. **(4 marks)** *November 2018*

27 Give any necessary and sufficient condition for a ring R to have no zero divisors. Justify your claim. **(4 marks)** *November 2018*

28 Show that cancellation law holds in a ring R if and only if R has no zero divisors. **(7 marks)** *November 2021*

29 Let n be a positive integer. Prove that : (a) The congruence class $[a]_n$ has a multiplicative inverse in $\mathbb{Z}_n$ if and only if $(a, n) = 1$. (b) A non zero element of $\mathbb{Z}_n$ is either has a multiplicative inverse or is a divisor of zero. **(5 marks)** *November 2023*

30 Show that any finite integral domain is a field. **(5 marks)** *November 2025*

Section C: 10 marks and above

Homomorphisms and kernels

31 Let G and G' be groups and let Φ be one to one function such that $\Phi(xy) = \Phi(x)\Phi(y)$. Prove that $\Phi[G]$ is a subgroup of G' and Φ provides an isomorphism of G with $\Phi[G']$. **(7 marks)** *November 2018*

The fundamental homomorphism theorem

32 State and prove Fundamental Homomorphism theorem. **(10 marks)** *November 2022*

Zero divisors, integral domains and fields

33 Establish a formula for computing the number of zero divisors in the ring $\mathbb{Z}_n$ by giving its proof. **(7 marks)** *November 2018*

34 Show that in the ring $\mathbb{Z}_n$, the divisors of 0 are precisely those elements that are not relatively prime to n also show that $\mathbb{Z}_p$ is a field. **(13 marks)** *November 2021*